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MILITARY SPECIFICATION

# TRAILER, FLAT BED; MINIMUM 15 TON PAYLOAD, MULTI-AXLE, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a multi-axle flat bed heavy hauler machinery trailer having a capacity to transport a 15-ton minimum payload up to 30 mph speed. In accordance with DOT 49CFR567, the trailer shall have a minimum 32,000 pounds Gross Vehicle Weight Rating (GVWR) at highway speeds. The trailer is of the cart type equipped with a tow eye suitable for towing behind pintle and drawbar equipped prime movers for earthmoving and construction jobs.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-C-13486 | - Cable, Special Purpose Electrical, Low-Tension, Heavy Duty Single Conductor and Multiconductor. |
| MIL-S-40626 | - Sign Kit. |
| MIL-E-52798 | - Enamel, Alkyd, Camouflage. |

FSC 2330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-209 | - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment. |
| MIL-STD-642 | - Identification Marking of Combat and Tactical Transport Vehicles. |
| MS75020 | - Connector, Plug, Electrical - 12 Contact, Intervehicular 28-Volt, Waterproof. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z35.1 - Industrial Accident Prevention Signs, Specifications for.

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA Standards.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A242 - High-Strength Low-Alloy Structural Steel, Specification for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

Welding Handbook.



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(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

DEPARTMENT OF LABOR (DOL)

29CFR1926 - Occupational Safety and Health Adminsistation (OSHA) Standards for Construction Jobs.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR393 - Federal Motor Carrier Safety Regulations (FMCSR).
49CFR567 - Certification.
49CFR571 - Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

FORGING INDUSTRY ASSOCIATION (FIA)

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, 55 Public Square, Cleveland, OH 44101.)

FRICTION MATERIALS STANDARDS INSTITUTE (FMS)

FMS Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Paramus, NJ 07652.)



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INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Application for copies should be addressed to Industrial Fasteners Institute, 1505 East Ohio Building, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW, Washington, DC 20005.)

REGULAR COMMON CARRIER CONFERENCE (RCCC)

RCCC Recommended Practices Manual.

(Application for copies should be addressed to the Regular Common Carrier Conference, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (T&RA)

Tire and Rim Association Yearbook, with Military Supplement.
TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)



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TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC., (TTMA)

TTMA Recommended Practices.
TTMA Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRDME-H, Fort Belvoir, VA 22060.)

## 3. REQUIREMENTS

3.1 Description. The flat bed trailer shall be of the commercial, low bed, machinery cart type having a front towbar with tow eye and centrally mounted multi-axle suspension under the bed. The trailer shall be the latest commercial production model of the contractor and shall have been marketed for a period of not less than 1 year (see 6.9), prior to opening of the technical proposal or bid, except for product improvements which have been incorporated on commercial units. The trailer shall be equipped with all components, furnished as standard and optional equipment whether specified herein or not, that are necessary to enable the trailer to function correctly and reliably in its intended application.

3.1.1 Weights and dimensions. The curb weight of the trailer shall include the weight of the chassis complete with all accessories and components. The Gross Vehicle Weight Ratings (GVWR), in accordance with DOT 49CFR567 and TTMA RP No. 39 (for highway speeds) 60 mph, shall be the curb weight of the trailer plus the weight of the equipment or material payload carried. Established components and vehicular ratings shall not be raised to conform to this specification. The physical characteristics of the trailer shall be as specified in Table I when the trailers are on a level surface, when the inflation of the tires are as recommended by the Tire and Rim Association for GVWR operation at highway speeds, and when the suspension systems are in the normal height position.



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Table I. Weights and Dimensions

| Description | Requirement |
|---|---|
| Curb Weight: Fixed Deck, Style F | Max. 8,400-lb. |
| Tilt Deck, Style T | Max. 8,800-lb. |
| DOT GVWR, Hwy Speed (60 mph) (96 km/hr) | Min. 32,000-lb. |
| Payload Capacity (Min. 30 mph) | Min. 30,000-lb. |
| GVW Distribution at Tow Eye | Max. 15% of GVW |
| Overall Length, OAL | Min. 280-in. |
| Overall Width, OAW | Max. 96-in. |
| Overall Height, OAH (Spare Tire Mounted) | Max. 78-in. |
| Ground Clearance, excluding Running Gear. | Min. 6-in. |
| Angle of Departure per SAE J894, Unloaded Level Condition: | |
| Fixed Deck, Style F | Min. 10-deg. |
| Tilt Deck, Style T | Min. 16-deg. |
| Tow Eye to Deck | Min. 70-in. |
| Tow Eye Height Ranges: | |
| - For Truck (Ref. SAE J849) | Max. 33-in. |
| - For CIM (Ref. SAE J749) | Min. 26-in. |
| - No. of Possible Positions | Min. 3-each |
| Trailer Bed: | |
| - Deck Height | Max. 39-in. |
| - Deck Length | Min. 216-in. |
| - For Tilt Deck, Angle of Deck Loading | Max. 17-deg. |
| - Load Capacity any 1-sq. ft. | Min. 8,000-lb. |

3.1.2 Compatibility. The trailers shall be compatible for towing behind trucks equipped with SAE J849 towing facilities and for towing behind construction equipment having SAE J749 towing drawbars, either with air and electrical or without air and electrical facilities available (see 6.12).



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3.1.3 Material. Material shall be as specified herein and shall be new and unused. Materials not specified shall be selected by the contractor as being compatible with the intended use in accordance with the engineering standards of the trailer and construction equipment industry and as set forth in the following industry standardization publications:

Society of Automotive Engineers (SAE).
Truck Trailer Manufacturers Association (TTMA).
Tire and Rim Association (TRA).
Friction Materials Standards Institute (FMS).
Industrial Fasteners Institute (IFI).
American Welding Society (AWS).
Forging Industry Association (FIA).
American Society for Testing and Materials (ASTM).
American National Standards Institute (ANSI).

Threaded parts shall be in accordance with SAE J370, SAE J371, and Industrial Fasteners Institute Standards.

3.1.3.1 Material design criteria. A calculated stress analysis of the trailer shall be required. The normal static stresses in the load-carrying members of the trailer shall not exceed 50 percent of the guaranteed yield point of the material when the trailer is loaded with the 15-ton payload specified in Table I. This analysis shall include calculations and diagrams for the following:

(a) The 15-ton payload GVW, load distribution diagram.
(b) The vertical shear diagram for the length of the trailer.
(c) The bending moment diagram for the length of the trailer.

3.2 Environmental conditions. The trailer shall be operational as specified herein in any ambient air temperature from plus 120° F to minus 40° F. When specified (see 6.2), the trailer shall be operational as specified herein in any ambient air temperature from plus 110° F to minus 65° F for arctic use.

3.3 Trailer performance. The trailer shall conform to the performance characteristics specified herein without malfunction, permanent deformation, or damage when (1) loaded with the rated payloads for the speeds specified (see Table I), (2) lubricated, serviced, and adjusted as specified in 3.6 through 3.6.2, and (3) towed by trucks and self-propelled construction equipment. The trailer performance shall be as follows:



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(a) Travel over smooth, primary highways at sustained speeds specified in Table I.
(b) Travel over unimproved roads, open fields, rolling hills and off-road terrain at speeds up to 20 mph.
(c) Negotiate mud up to 12-inches deep and standard roadway V-ditches up to 36-inches wide and 18-inches deep.
(d) Negotiate side slopes up to 10 percent.
(e) Have sufficient articulation to permit the trailer to roll over obstacles with any two wheels without loss of tire-to-surface contact on any wheel; a minimum of 6-inch-high obstacle for the fixed deck, Style F, and a minimum of 3-inch-high obstacle for the tilt deck, Style T.
(f) Trail the prime mover without bottoming, weaving, or side sway and without exceeding the DOT FMCSR towed vehicle tracking allowance (see 3.11).
(g) Permit turns up to a minimum angle of 60 degrees from straight ahead to the right and to the left without interference or damage of truck or trailer; permit pivot turns about the center of the trailer suspension.
(h) Provide braking ability to decelerate, control, and stop in compliance with DOT FMVSS, DOT FMCSR, and the California Vehicle Code and have brake system integrity in accordance with TTMA RP No. 12 and RP No. 44.
(i) When coupled to a prime mover, withstand loading and unloading of construction equipment, weighing at least 30,000 pounds and having twin concentrated load areas of skids, track laying crawlers, or wheels; accomplish loading and unloading under conditions where the side sloping contour of the terrain results in at least 6-inches difference in elevation between the rear right and left corners of the deck and the ground.
(j) Permit coupling and uncoupling of the trailer, to prime movers by one man without resorting to means other than the front support leg.
(k) Provide front support leg capable of withstanding the following:

(1) Static and dynamic forces due to the proportionate weight of the fully loaded trailer;
(2) Impact forces during coupling and uncoupling operations, including a free fall drop of at least 4-inches.

3.4 Trailer specifications. The characteristics of the trailer shall be in accordance with Table II.



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Table II. Trailer Specifications

| Feature | Requirement |
|---|---|
| (a) Towbar and Frame | Per DOT FMCSR 393.71(h); frame of either ASTM A242 or SAE J410, HSLA steel having minimum 50,000-psi guaranteed yield point (YP); stress maximum 50 percent of YP; continuous longitudinal framing members; withstand towbar pull equivalent to 100 percent of trailer GVW with 30,000-lb. payload; lockable accessory and tool compartment(s) with cover(s) furnished in towbar area to carry tools, come-a-longs, chains, and blocks; unless otherwise specified, min. compartment size manufacturer's standard size; when and as specified (see 6.2), volume not less than 4200 cu. in. eyebolts or footman loops provided to secure loose items; drain holes provided; spare tire mount provided and include padlock securing means. |
| (1) Tow Eye | Per DOT FMCSR 393.70(c) and SAE J847; alloy steel eye with case hardened wear surface; multiposition vertical range of adjustment at least 26-inches, per SAE J749, thru 33-inches, per SAE J849; center of tow eye within maximum 1/4-inch of longitudinal centerline of the trailer. |
| (2) Safety Chains | Per DOT FMCSR 393.70(f) and SAE J697; 2 each, extend minimum 36-inches forward of tow eye center; safety type chain hook on truck-ends of chains. |
| (3) Front Support Leg | Min. 20,000 lb. rated load capacity, semitrailer type; provide minimum 15-inch vertical adjustment; enclosed raising and lowering mechanism protected from dust and dirt; all-weather-type lubricant in mechanism; leg either retract or fold up to provide minimum 12-inch ground clearance, when level; folding crank handle actuation; self-alining mud pad type foot; if removable, foot captive chained or cabled to trailer and stowed in tow bar box. |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| (b) Platform | Size per Table I; full width crossmembers integrally welded to frame rails and platform frame; replaceable, weather-resistant-treated, wood plank flooring having minimum 2-inch nominal thickness; wood decking 1/4 secured with plated recessed bolts and nuts, clamping undrilled framing members; wheel covers of minimum 5/16 inch thick, nonskid pattern, steel plates, when furnished: Unless otherwise specified, platform equipped with 6-ea. stake pockets, 3-ea. symmetrically spaced along each side, to accept 4-inch square standard timber stakes; when specified (see 6.2), platform equipped with 6-ea. round pockets to accept 3-in. diameter standard weight (ASTM 120) pipe stakes; provide stake or pipe pockets flush with top of platform and partially closed at the bottom, sufficiently to both hold the stakes and permit drainage of rain; front equipment stop member maximum 4 inch height; unless otherwise specified, provide a tilt deck platform style; when specified (see 6.2), provide a fixed deck platform style. |
| (1) Style F, Fixed Deck | Integral welded frame and platform with rear beavertail and fold-over, adjustable, rear ramps, one on each side; beavertail maximum 60-inch horizontal length. |
| (2) Style T, Tilt Deck | Separate frame and pivoting platform; platform provide tilted, loading position and level, travel position; platform balanced about the center hinge pivot to permit the weight of one person (maximum at 175 pounds) to tilt platform into positions; tilt dampening control provided by double-acting hydraulic cylinder(s); |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| | platform equipped with manual or automatic lock securing platform by positive means to drawbar without chatter and without side-shift in excess of 1 inch total movement; platform lock, in conjuction with the platform pivot pins, comply with the forces and intent of DOT FMCSR 49CFR393.100(e); platform lock release accomplished manually, without use of tools; tilt bed beavertail permissible only if horizontal distance from beavertail break to rear of trailer 36 inches or less. |
| (3) D-Rings | Minimum 14 each; 5 each on each side; 2 each on front, and 2 each on rear; each D-ring and installation withstand minimum 10,000 pound pull in any direction. |
| (4) Outriggers | When specified (see 6.2), outriggers, with 2 X 12 lumber planks, furnished on each side of platform; outrigger extension arms of hinged swing-out type with locks; minimum 10 each, 5 each per side. |
| (5) Skid Plate | Rear underside of platform equipped with steel skid plate to reduce ground resistance across ditches and off-road terrain; plate minimum 94 inches wide and minimum 36 inches forward from rear cross-member or the length of the beavertail, if furnished. |
| (c) Loading Winch | When and as specified (see 6.2), furnish either a hand-operated or an electric-power-operated loading winch to pull inoperative equipment onto the trailer. |
| (1) Winch | Winch of single drum, 2-speed type having a min. 4000-lb. capacity, 1st layer line-pull rating; gears per AGME Standards; provide both handbrake and safety ratchet |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| | and pawl lock; rotation of drum in the winch-in direction to result in ratcheting of pawl;<br>min. drum diameter 7-times nominal winch wire rope diameter; hand-operated winch geared to deliver rated capacity line pull and not to exceed 40 ft. lbs. of hand crank effort at the handle;<br>electric power-operated winch geared to deliver rated capacity line pull at 12-volt, negative ground, dc battery power from the prime mover;<br>electric power for winch-in, hold, and winch-out of line;<br>provide remote winch control having leads harness a minimum of 10 feet in length;<br>momentary operator switches controlling electrical relays;<br>prime movers 12-V power received from Auxiliary, BLUE, No. 7 hot terminal and Ground, WHITE, No. 1 terminal of TTMA RP No. 40 trailer receptacle;<br>winch motor wired with 12 GA hot wire and 8 GA ground wire to TTMA receptacle;<br>winch located on front of tilt deck on or forward of equipment stop member;<br>winch fastened to trailer using minimum Grade 8 bolts, Class C nuts, and integral or hardened steel washer in matching sets per IFI and SAE;<br>Winch equipped with fairlead rollers to provide wire guidance to the rear of the trailer;<br>Two horizontal rollers and two vertical rollers;<br>lube fittings provided for rollers. |
| (2) Winch Line Assembly | Provide min. of 53 feet of winch wire rope and chain assembly installed on the winch drum, consisting of trailer deck length |



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Table II. Trailer Specifications (Cont'd)

| Feature | | Requirement |
|---|---|---|
| | | plus an additional min. 35 feet reach rearward from the rear of the trailer; on the working end of the winch wire rope, provide a swivel and a min. 3 foot length of chain with chain grab hook to attach winch line to equipment; min. winch cable size, 3/8-in. dia., 6 by 19, min., IWRC, very high strength (VHS) winch wire rope having each wire strand individually formed, providing max. exterior wear surface area; all components to have a working load rating in excess of the winch rating and a breaking strength that has a minimum safety factor of 5:1 above the working load. |
| (d) | Tandem Suspension System | Tandem, rated capacity at least equal to the maximum load imposed at the ground; withstand both highway and off-road travel; suspension installation in correct alinement with tow eye and trailer centerline, per both suspension and axle manufacturers; sufficient articulation to maintain equal loading on each wheel per 3.3(e), Trailer Performance; provide positive control over brake hop and axle rotation during braking; provide elastomer stops or bumpers to prevent metal-to-metal contact at maximum wheel compression position; Min. 48-in. tandem wheelbase. |
| (e) | Axles | Per DOT FMVSS and TTMA RP No. 39, provide 2-each axles providing min. 34,000-lb. DOT tandem GAWR, each axle having min. 17,000-lb. rating at hubs; spindles of alloy steel hardened and ground; standardized tapered roller bearings with cups, bearings, and cones in matched sets, all from the same bearing manufacturer on each trailer; oil lube with seals per TTMA RP No. 43 having visual window caps; axle hubs per SAE J694, 10 stud, 8-3/4 inch BC. |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| (1) Hubodometer | 1 each driveless, tamperproof, self-lubricating type for converting wheel revolutions into accurate mileage readings; visual recording counter with numbers in the upright position; minimum 7-digit type recording a minimum 999,999.9 miles; corrosion-resistant housing sealed against moisture, oil, grease, dust, and climatic conditions; hubodometer located on rearmost curbside axle hub of fixed deck, Style F trailers and front curbside axle hub of tilt deck, Style T trailers. |
| (2) Tires and Wheels | All components per DOT FMVSS and TRA Yearbook; heavy-duty, disc wheel and tire assemblies; rim and wheel disc fabricated by same manufacturer or as approved by rim manufacturer; tires minimum 100 level quality; cold inflation matched to loading per TRA; metal valve caps with elastomer seals; tires, wheels, hub and drum or disc, all inherently in-balance and not beyond industry tolerances to prevent wheel tramp, drumming, or excessive tire wear; wheel splash protection per SAE J682.<br>8 each low-platform trailer dual tandems having minimum size 10.00 - 15 TR, minimum LR F (12 PR); Tube type; cold inflation, maximum 80 psi; radial-type inner tube furnished regardless of type of tire; flaps per TRA.<br>1-ea., identical, inflated spare tire and wheel assembly furnished and towbar area mounted with each trailer. |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| (h) Brake System | Per DOT FMVSS, DOT FMCSR, and California Vehicle Code; straight air service and emergency-parking system; withstand high speed, off-road, and mountainous travel and creek bed immersion usage; provide priority to brake system over any auxiliary air system use. |
| (1) Gladhands | Per SAE J318; yellow for service (control) and red for emergency (supply); spring loaded gladhand dust covers; minimum 36 inch hose (see "(7)-Plumbing") extension forward of tow eye center, compatible with SAE J849; clips or dummy gladhands to hold lines doubled back on trailer drawbar; ID tags at tow bar fitting. |
| (2) Relay Valves | High speed type. |
| (3) Air Reservoirs | Minimum 2 each per DOT and SAE J10; 1-each service and 1-each parking release, both sized per DOT and component manufacturer's recommendations; automatic drain valves on each reservoir; no component lower than bottom of frame rails. |
| (4) 121 Antilock System | When legally required, per DOT and RCCC RP's; hermetically sealed and water immersionproof electrical components; minimum 80 percent of anti-lock systems electrical cable lengths covered or encased by grounded metal (including frame members) or system filtered to prevent external radio and electromagnetic interference; TTMA RP No. 42 indicator located near the axle; indicator of the flush or the in-line lamp, of same preferred types as furnished for the trailer's marker lamps, with bulbs and red lenses; to prevent excess circuit overloading, delete 1-each bulb on double bulb lamp |



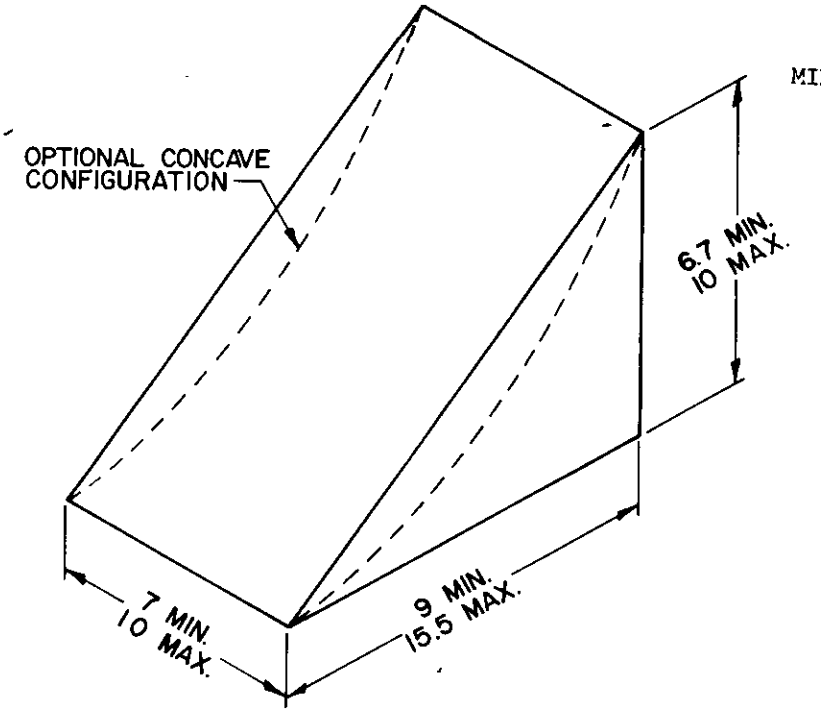
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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| | types; axle tube holes sealed waterproof by grommets and silicone RTV sealant around sensor wires; |
| (5) Foundation Brakes | Disc type or wedge or S-cam type with removable dust shields; components protected from damage, located above the bottom of the brake disc or brake drum; discs, wedges, cams, and roller pins, as a minimum, hard-chrome plated and silicone high temperature compound on bearing surfaces, effectively preventing corrosion malfunction; parking release system for towing by prime movers without airbrake system; lining nonfade mountainous type, heat and water resistant; standard size listed in FMSI Automotive Data Book. |
| (6) Brake Actuators | Unless otherwise specified, provide actuators of the heavy-duty spring type having environmental protection covers, boots, and seals; elastomer coated mainsprings capable of holding all wheels locked on grades up to minimum 30 percent; mainspring accessible and capable of being manually caged by tools or means supplied on each trailer, to permit trailer towing by crawler tractors and other construction equipment not having air brakes; self-adjusters of enclosed mechanism type for each wheel, providing correct compensation for lining wear without lockup or unbalance; manual means to adjust in case of malfunction; when specified (see 6.2), provide brake actuators without heavy-duty emergency parking mainspring; however, for parking brakes, provide at least 4 each, captive chained, lightweight chock blocks per DOL OSHA and Figure 1. |



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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT-11.5 LB.

CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

**FIGURE 1. CHOCK BLOCK ASSEMBLY.**



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| (7) Plumbing | All hose type with reusable, SAE J516, 100R5 swivel fittings using SAE J1402, Type D, Class II hose; routing within protection of framing members; elastomer cushioned clamps and grommets secure and support hose; identify hose line function adjacent to each fitting with weatherproof means; minimum 3 inches of slack at relay valves; all fittings per SAE; cast iron pipe fittings not acceptable. |
| (i) Vehicle Electrical System (12-Volt) | Per DOT, California Vehicle Code, SAE Handbook, TTMA RP No. 9, TTMA RP No. 14, and RP Manual; heavy-duty waterproof system providing protection from physical damage, shock, vibration, corrosion, and environment; system 12-volt potential with negative ground per SAE J538. |
| (1) Front Plug and Cable | Compatible with SAE J849; plug per TTMA RP No. 40 with cable per SAE J1067 having minimum 168-strand flexible ground wire; plug extend minimum 36 inches forward of tow eye center; clip to hold cable doubled back on trailer towbar; weatherproof, towbar mounted connection box with nut and stud terminals and SAE J553 circuit breakers protecting all live circuits; no fuses. |
| (2) Wiring | Per SAE J163, and SAE J555, and TTMA RP No. 14; wire identification per SAE J560, or optionally, SAE J821; multiple circuits combined into waterproof harness; wiring supported at maximum 36 inch spacing with elastomer ties and secured mounts or straps to prevent whipping, chafing, and mechanical or fatigue damage; no tautness or looping of wire; interior of splicing devices waterproofed with silicone RTV sealant; |



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Table II. Trailer Specifications (Cont'd)

| Feature | | Requirement |
|---|---|---|
| | | wiring cable per RCCC RP - 110; terminals per RCCC RP - 112; terminals plated or dip soldered and means of strain relief and heat shrinkable insulation sleeves provided; main junction blocks of nut and stud type having silicone or electrical varnish coating after assembly; wire-to-wire terminals secured to vehicle with plastic spring clips. |
| (3) | Marker Lamps | Flush mount or in-line protective lens surface mount types per RCCC RP No. 108; RP No. 108/1 or RP No. 108/3 protective lens of the preferred lamp types; in-line type lamps recessed within the main frame flanges to prevent side-swipe damage. |
| (4) | Rear Combination Lamps | Per SAE J759, SAE - IST, and TTMA RP No. 14, flush=mounted elastomer housing with bulb and snap-on or gasketed lens or sealed unit with eleastomer ring; round 4 inch diameter size installed in TTMA standard hole; double-filament configuration with 3-wire pigtail, including ground to frame; waterproof and shock resistant; total 4-each; 2-each outboard turn and clearance and 2-each inboard stop and tail functions; installed in rear crossmember having vertical gussets furnished to prevent flange crushing by loading and unloading equipment. |
| (5) | Lamp Bulbs and Lenses | Hermetically sealed units or fleet service heavy-duty bulbs (RCCC Advisory AV 1-2); bulbs mounted in brass, vibration protected sockets with ceramic or plastic insulators and stainless steel or elastomer bulb tension devices; lenses of polycarbonate plastic. |



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Table II. Trailer Specifications (Cont'd)

| Feature | Requirement |
|---|---|
| (6) Reflectors | Min. 8-each with single or twin metal fasteners securing reflectors to the trailer; install 2-each additional reflectors facing forward for use when parked alone. |
| (7) Wiring Diagram | Metal plate, showing wiring, furnished per TTMA Technical Bulletin No. 65 and "Metal Name Plate Industry Standards and Practices" per NAME. |

3.5 Military vehicle electrical system. When specified (see 6.2), the trailer shall be additionally equipped with a military compatible, 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-way plug, the trailer shall be equipped with the following 12-way military inter-vehicle components:

(a) Military 12-Contact Plug - MS75020-1.
(b) Military Intervehicular Cable - MIL-C-13486/12.

The military plug and cable assembly shall be the same length as the TTMA plug and cable assembly and both shall be labeled with identification plates for 24-volts and 12 volts respectively, negative ground. The military 24-volt intervehicular cable shall be routed into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with Table III. The resistors shall be exposed to ambient air for cooling and shall be protected from damage, road debris, and accidental burning of operating personnel.



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Table III. Military-Commercial Connections

| Function | Mil. Plug Terminal | Connection Instructions | Comm. TTMA Wiring Ident. |
|---|---|---|---|
| Blackout Marker | A | No. | - |
| Ser. Left Turn & Stop | B | Connect to - | YEL (3) |
| Blackout Marker | C | No. | - |
| Ground | D | Connect to - | WHT (1) |
| Ser. Tail & Marker | E | Connect to - | BLK (2) |
| | | & Connect to - | BRN (6) |
| Blackout Stop | F | No. | - |
| Blackout Marker | H | No. | - |
| Ser. Right Turn & Stop | J | Connect to - | GRN (5) |
| Aux. Power (Hot) | K | No. | BLU (7) |
| Ground | L | Ground to Frame | - |
| Electric Brake | M | No. | - |
| Electric Brake | N | No. | - |

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.6 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. The lubrication intervals shall be as recommended by the component manufacturers and in accordance with SAE J752. A metal lubrication chart, conforming to SAE J753, shall be furnished on the towbar or number one crossmember. The trailer shall operate as specified herein when serviced with military lubricants (see 6.5).

3.6.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534, threaded types, and shall be accessible to a 15-ounce grease gun. Fittings shall be located in a protected position. Where oil holes are applicable, they shall have spring-loaded covers.



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3.6.2 Servicing and adjustment. Prior to delivery, the contractor shall service and adjust each trailer for correct operation, including checking and adjustment of landing leg and of the brake system, continuity check of the electrical system, torque of all wheel lug nuts, inflation of tires, and complete lubrication of the trailer with the types and grades of lubricants recommended.

3.7 Ease of maintenance. Components of the trailer chassis shall be accessible for servicing, repair, and replacement. The adjustment and replacement of components shall be accomplished with minimum disturbance to adjacent components. Ease of maintenance provisions shall incorporate features insuring operating clearances in accordance with SAE J925, and facilitate maintenance and service operations.

3.8 Treatment and painting. The trailer chassis shall be cleaned, treated, primed, and painted in accordance with the manufacturer's standard practice. Unless otherwise specified (see 6.2), the finishing coat shall be in accordance with MIL-E-52798, Forest Green color, and markings shall be of black color. The inaccessible areas of the trailer shall be treated with anti-rust coating material in accordance with TTMA Technical Bulletin No. 56.

3.9 Marking. The U.S. Army markings shall be as specified in the solicitation and in MIL-STD-642 (see 6.8), color black. The tire pressure shall be marked near the tires and the greatest "GVW @ __ MPH" shall be marked on the road side of the towbar.

3.10 Slinging and tiedown provisions. Permanently affixed slinging components shall be furnished in accordance with MIL-STD-209 to lift the trailer at its curb weight in its normal travel position, through use of the trailer's D-rings and tow eye. Slinging components shall also be used as tie downs to permit fastening of the trailer to the floor or deck of another transportation vehicle. The contractor's slinging and tie down procedures shall be included in the trailer's manuals.

3.11 Safety. The safety characteristics of the trailer shall meet the following requirements:

(a) Conform to all requirements of the DOT Motor Carrier Safety Regulations pertaining to large trailers, 49CFR390 and 393.
(b) Conform to trailer requirements of the Federal Motor Vehicle Safety Standards 49CFR571.
(c) Conform to the California Vehicle Code requirements for trailers.
(d) Comply with OSHA requirements, 29CFR1926, for vehicles on construction jobs.



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(e) Step surfaces shall provide a nonskid footing and shall comply with SAE J185.
(f) Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be eliminated.
(g) The trailer shall provide for operation by personnel in accordance with SAE J833 and SAE J925.

3.12 Government-furnished equipment (GFE). The contractor shall provide fasteners and mount the following GFE on each trailer (see 6.7):

(a) Case, Maintenance and Operator's Manuals, NSN 7520-00-559-9618.
(b) Materiel Fielding Plan.
(c) Prescribed Load List (PLL) and Authorized Stockage List (ASL).
(d) Vehicle classification sign kit (reference MIL-S-40626, NSN 9905-00-565-6267).

3.13 Basic issue items and commercial manuals. The contractor shall overpack each trailer with the items specified in the Basic Issue Items List (BIIL) (see 6.6) and tire lug wrench and other special trailer tools required for trailer operation and trailer operator maintenance. Two copies of the Government approved operator, lube, maintenance and overhaul, and parts manuals shall also be provided and overpacked with each trailer.

3.14 Identification and instruction plates. The contractor shall provide all identification and instruction plates normally furnished and as specified herein. Identification and instruction plates shall be metal. Plates and their installation shall conform to the requirements of "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers (NAME).

3.14.1 Identification plates. The trailer shall be identified in accordance with DOT, TTMA, and the trailer manufacturer. In accordance with TTMA RP No. 39, the trailer's VIN plate shall include, in addition to the DOT GVWR, the maximum GVW's and payload allowance for towing speed related maximums of 30 mph and 20 mph. The safety, warning, caution, and notice plates shall be in accordance wtih OSHA and ANSI Z35.1, describing any important or special procedures to be followed by users in operating or servicing the trailer. A shipping data plate shall be furnished showing the silhouette of the trailer in the transport position, indicating the trailer's center of gravity at curb weight, and identifying the tow eye and D-rings and recommended capacity to be used in lifting and tie-down for shipment.



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3.15 Technical Information Package. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2), depicting the trailer and its components exactly as it will be furnished. The Technical Information Package shall be submitted with all bids and proposals.

3.16 Certification. Prior to the first article evaluation (see 4.2), the contractor shall provide certification that the trailer conforms to the following to the extent specified herein:

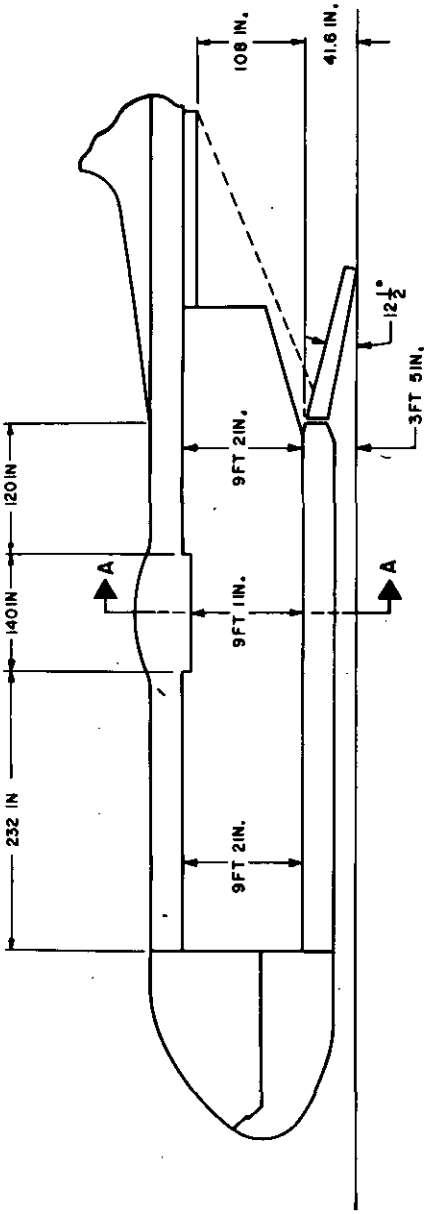
(a) DOT FMVSS.
(b) DOT FMCSR.
(c) California Vehicle Code.
(d) DOL OSHA Regulations.
(e) Industry Standardization Document Compliance.

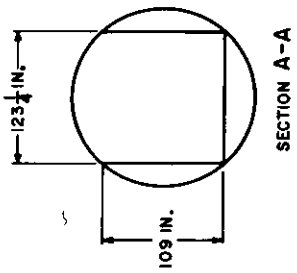
3.17 Transportability. At curb weight, the trailer shall be transportable in the following:

(a) In C-130, C141, and C-5A aircraft as shown in Figures 2, 3, and 4 without trailer contact with the aircraft loading ramps.
(b) On a flatcar to comply with the requirements of the "Rules Governing the Loading of Commodities on Open Top Cars" of the Association of American Railroads.



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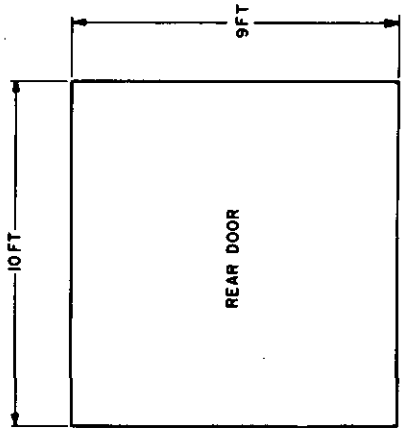
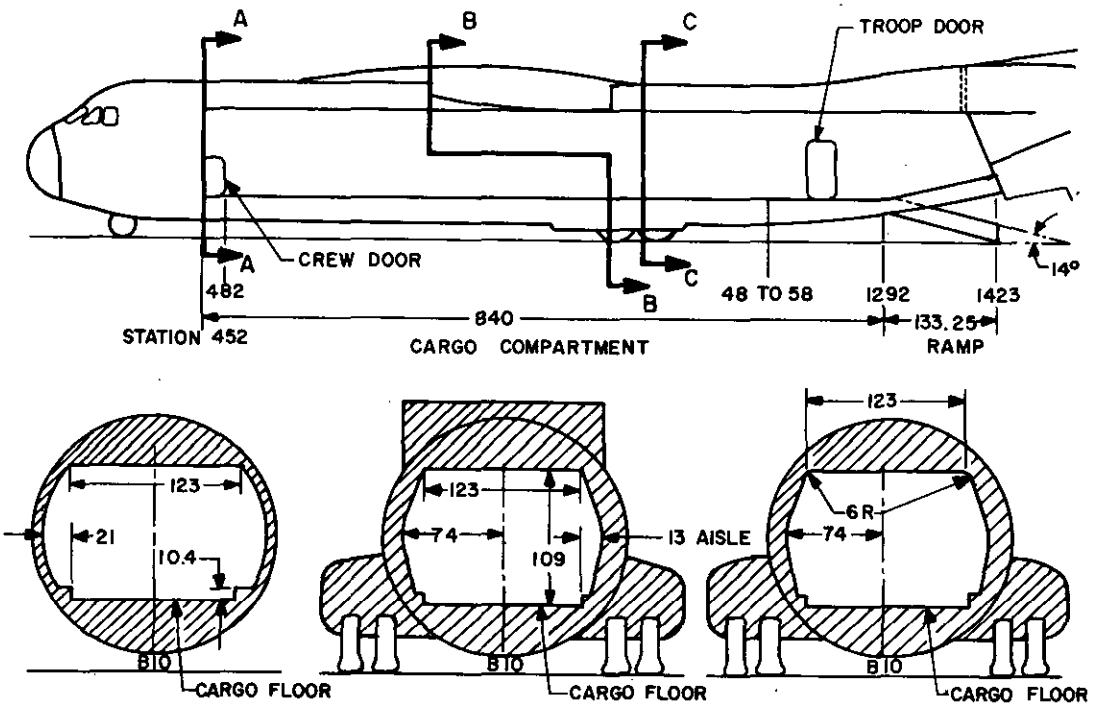




FIGURE 2.CARGO COMPARTMENT PROFILE (C-130). X-195



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TYPICAL FROM STA 452.10 1108 EXCEPT AS NOTED — SECTION A-A

TYPICAL AT STA 734.7 AND 958 — SECTION B-B

TYPICAL AT STA 998 AND 1058 — SECTION C-C

CARGO COMPARTMENT DIMENSIONS AND CONTOURS C-141 AIRCRAFT

MAIN LOADING DOOR:
WIDTH - 123 (APPROXIMATELY)
HEIGHT- 109 (APPROXIMATELY)

NOTES:

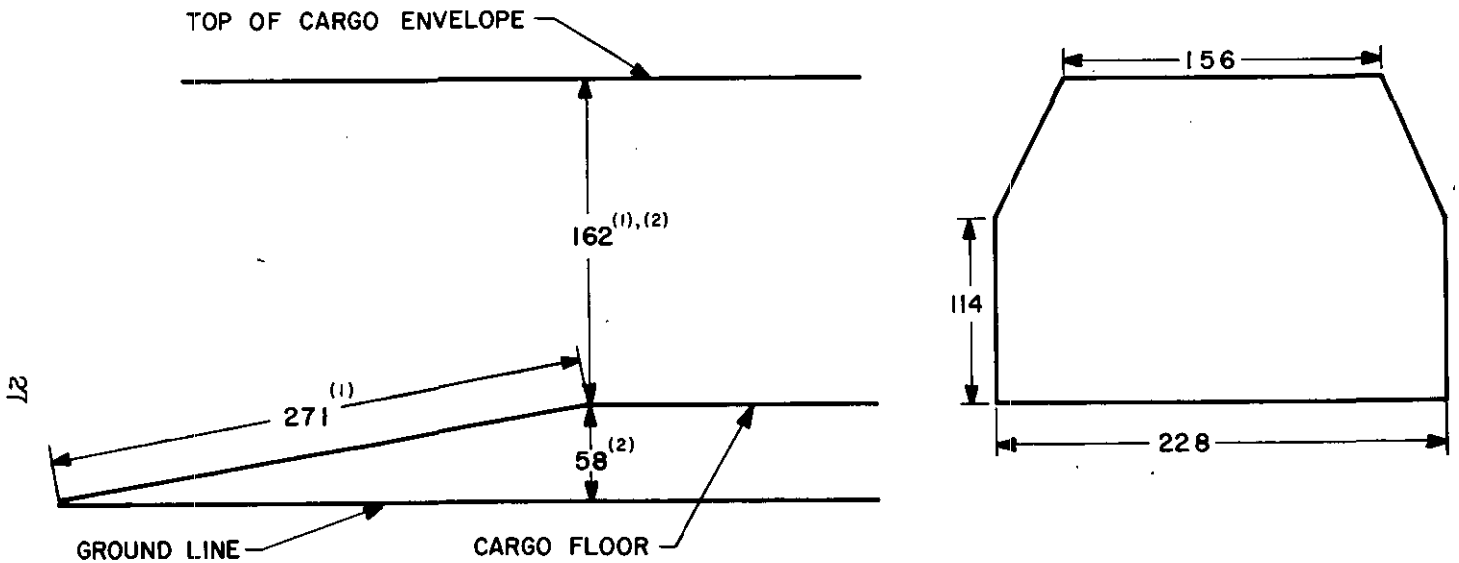
1. ALL DIMENSIONS ARE IN INCHES
2. THE HEIGHT AND WIDTH OF THE COMPARTMENT ARE REDUCED 2 INCHES ACROSS THE UPPER OUTSIDE CORNERS OF THE COMPARTMENT AT FUS STA 1198 AND 1238 BY THE TROOP DOOR TRACKS
3. WHEN THE ROLLER CONVEYORS ARE TURNED UPRIGHT, USABLE HEIGHT WILL BE REDUCED 1.5 INCHES

FIGURE 3. C-141 Cargo envelope. X-3170



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FORWARD CARGO OPENING C-5A
FORWARD KNEEL POSITION




NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. REFERENCE: AMCP 706-130, ENGINEERING DESIGN HANDBOOK, DECEMBER 1967.
3. REFERENCE: USATECOM PROJECT NO 4-ES-065-050-004, SERVICE TEST OF C-5A AIRCRAFT, FIRST PARTIAL REPORT BY H.H JOHNSON, JR., CPT., JANUARY 1971.

FIGURE 4. C-5A. cargo envelope.

X-3171A

MIL-T-5284B(ME)

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3.18 Vehicle weight classification. The contractor shall complete and submit the Government-furnished vehicle weight classification forms and install the vehicle weight classification number kit (ref MIL-S-40626), on the trailer (see 6.10). The kit shall be located on the front of the trailer in a location approved by the contracting officer. The forms shall be submitted for DOT defined highway speeds 60 mph and for 30 mph and 20 mph maximum trailer GVW's.

3.19 Workmanship. All workmanship on each trailer shall be in accordance with engineering, manufacturing, and production standards of the trailer industry. The trailer chassis shall be free from workmanship deficiencies that could impair the operation of serviceability of the trailer chassis. All parts, components, and assemblies of the trailer chassis, including bearings, seals, machinery, stampings, welded parts, castings, forgings, and machined work, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges and corners are functional. Welding shall be in accordance with the AWS Welding Handbook.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

4.2 First article evaluation. The first-produced trailer shall be compared with this document, the contractor's completed Technical Information Package, the published trailer and component literature, and the contractor's Bill of Materials and Assembly Build-up Sheet, to verify conformance to this specification. In addition, the trailer shall be subjected to a demonstration as specified herein. Nonconformance of the trailer to any requirement contained herein shall be cause for rejection of the trailer.



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4.2.1 Stress analysis and static load capacity. A stress analysis for the trailer shall be calculated and furnished, containing load, shear, and moment diagrams. The stress analysis of the trailer shall demonstrate that the maximum weight of the specified GVWR's and payloads (see 3.1.1, Table I) shall not produce a stress greater than 50 percent of the yield point of the material of any member of the frame when the payload is applied over the most critical 8 linear feet of the bed area. Support the trailer on the suspension frame points and the center of the drawbar towing eye. In the most critical 8 linear feet of the flatbed loading area, preload to 1.75 times the specified payload to relieve stresses, then unload. Establish and record the dimensional relationship of trailer points at the lowest outside edges of the mainframe members, below the load bed, on both the roadside and the curbside as follows:

Points A - Frame height at drawbar-to-bed junction.
Points B - Frame height just ahead of end of frame.
Points C - Frame height midway between Points A and B.

Impose a test load of 1.75 times the specified payload and record all 6 points. Unload and record all 6 points again. Compare the first set of recorded point measurements with the third set of recorded measurements. If permanent deformation of any member of the trailer has occurred, the indicated trailer payload rating shall not be acceptable.

4.2.2 Road demonstration. Prior to the road demonstration, the trailer shall be run-in and serviced. Tow the trailer as specified in Table IV for a total distance of not less than 130 miles. Unless otherwise specified herein, loading of the trailer shall be with the minimum 30,000 pounds payload specified in Table I, evenly distributed to provide the weight distribution specified in Table I. During the test, all trailer lights shall be illuminated, turn signals operated on turns, and hazard warning operated during stops.

Table IV. Road Demonstration

| Course | Mileage and Speeds | Loading |
|---|---|---|
| Primary highway | 40 mi at variable speeds up to 30 mph | Loaded |
| Primary highway | 40 mi at variable speeds up to 60 mph | Empty |
| Unpaved roads | 20 mi at variable speeds up to 20 mph | Loaded |
| Unpaved roads | 20 mi at variable speeds up to 20 mph | Empty |
| Cross-country | 5 mi at variable speeds up to 20 mph | Loaded |
| Cross-country | 5 mi at variable speeds up to 20 mph | Empty |



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During the road demonstration and while hauling maximum payloads, perform the following:

(a) Tow the trailer on side slopes up to 10 percent, with the curb side downhill and repeat with the road side downhill.
(b) Tow the trailer over the specified suspension articulation obstacles such that all left and right combinations of the wheels have simultaneously passed over two obstacles. Observe tire-to-surface contact, evidence of binding or interference and, after returning the suspension to normal, measure the diagonals and parallels for evidence of misalinement.
(c) Tow the trailer for not less than 100 feet through mud at least 12 inches deep. Any prime mover may be used for this phase of test. Back the trailer through the same course.
(d) Tow the trailer a total of 10 times over a V-ditch (standard Bureau of Public Roads highway type) not less than 36 inches wide and 18 inches deep. Negotiate the ditch both perpendicular and at an angle of 45 degrees to the direction of travel. Inspect rear skid plate for excessive permanent deformation.
(e) Measure and record the DOT FMCSR deviation or off tracking of trailer with respect to the prime mover, when the trailer is towed straight ahead on a paved surface.
(f) Jackknife the trailer at angles of at least 60 degrees from straight ahead both to the left and to the right.
(g) Turn the prime mover, towing the trailer, in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction.
(h) Shift the payload so that the trailer is balanced on its suspension system and no load is any longer on the front towbar. Using wood blocking or other means, block out the suspension articulation to the maximum extent possible. Pivot turn the loaded trailer about the center of the trailer suspension for at least 270 degrees to the left and repeat for at least 270 degrees to the right.

Failure to complete the demonstration, inability to operate and function as specified herein, or nonconformance shall constitute failure of this demonstration.

4.2.3 Brake demonstration. After suitable brake system break-in, weigh and record both the prime mover and the trailer with rated payload, (see Table I). The course shall be a hard, paved surface. Perform the following braking:



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(a) Demonstrate conformance to DOT FMVSS and DOT FMCSR.
(b) Demonstrate conformance to the California Vehicle Code.
(c) Test the brake system in accordance with TTMA RP No. 12 and, when legally required, TTMA RP No. 44.
(d) Demonstrate towing by vehicle, without airbrakes connected.
(e) When the trailer is equipped with chock blocks, disconnect the chock block captive chain and tow the trailer over the chock blocks, one at a time; inspect for any damage.

Nonconformance shall constitute failure of this demonstration.

4.2.4 Loading demonstration. Loading and unloading of the trailer shall be accomplished by one man.

(1) Park the trailer, coupled to the prime mover, on a level paved surface. Manually drop the fixed deck loading ramps or unlatch the tilt deck and tilt the deck from the travel to the loading position. Load the trailer with an item of construction or earthmoving equipment, tracked or wheeled, weighing at least 30,000 pounds. Drive the equipment to the trailer's fixed deck breakover point or tilt deck balance point and inch forward to lower the equipment to travel position. Inch the equipment rearward to raise to the loading position. Repeat this fore-and-aft inching movement of the equipment for a total of 5 cycles, while observing ramp movement or tilt deck action. Finally, drive the equipment forward and park the equipment.
(2) Unload and reload the equipment on actual or simulated side sloping terrain that produces a racking of the deck as specified in 3.3(i). Measure and inspect trailer.
(3) Lower the landing gear and uncouple the loaded trailer from the prime mover.
(4) Raise the front end of the trailer to a height necessary for the bottom of the landing gear pad to clear the paved surface by at least 4 inches vertical height. Rapidly remove all support at the front of the trailer, allowing the landing gear pad to strike the paved surface from the free fall height of 4 inches. Repeat landing gear impacts for a total of three free falls. Inspect landing gear and trailer.
(5) Operate the landing gear from fully retracted to fully raised not less than five times. Match lunette height to pintle and recouple prime mover and trailer. Unload the trailer and manually tilt the deck to the travel position or fold up the loading ramps.



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(6) Unless not required, extend deck extension arms and install the extension plank assemblies on each side of the deck. Repeat the loading, tilt cycling and landing gear cycling and unload in accordance with (1), (3), and (5). During the tilt cycling drive the equipment onto both extension planks and observe performance.

Nonconformance shall constitute failure of this demonstration.

4.2.5 Slinging and tiedown demonstrations. Demonstrate the lifting attachment capability and method and the tiedown attachment capability and method by applying the applicable loading forces as specified in MIL-STD-209 for the period of time required. Damage, permanent deformation, or nonconformance to 3.10 shall constitute failure of this demonstration.

4.2.6 Transportability evaluation. When specified (see 6.2), the first article will be subjected by the Government to a First Article Transportability Evaluation. This evaluation is to be performed by the Government at a Government selected site and will last no longer than 60 days, commencing with the delivery of the vehicle to the selected site. The Transportability Evaluation shall include loading, and securing the vehicle in aircraft as specified in 3.17. Nonconformance to 3.17 shall constitute failure of this evaluation.

4.3 First article demonstration report. The contractor shall prepare and submit the "First Article Demonstration Report" to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed, deficiencies encountered, and actions taken to correct the deficiencies.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete trailer and all items issued with the trailer shall be suitably preserved and packed for retail distribution directly to a using customer and in a manner to assure correct acceptance and safe delivery to the destination at lowest rating in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitor, or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, pack compartmentalization, or placing of the individual item(s) in a carton. Each reservoir shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail,



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and any other servicing required prior to placing the trailer in operation. Marking shall reflect, as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings on each unpacked item or shipping container. The marking shall be applied with a waterproof material, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The trailer is intended for use as a transporter of earthmoving and construction equipment and of construction materials when towed by both wheeled and track laying prime movers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When special winterized trailers are required and are intended for arctic use to at least minus 65° F ambient temperature (see 3.2).
(c) When a tool compartment, having a size other than the manufacturers standard size, is required (see 3.4, Table II, (a)).
(d) When stake pockets along the sides of the bed are required to accept 3-inch diameter pipe stakes (see 3.4, Table II, (b)).
(e) When a fixed deck platform style is required in lieu of the tilt deck platform style (see 3.4, Table II, (b)).
(f) When external outriggers on each side of the trailer are required to furnish support for oversize width payloads (see 3.4, Table II, (b), (4)).
(g) When chock blocks are required in lieu of fail-safe spring brake actuators (see 3.4, Table II, (h), (6)).
(h) When the additional military vehicle electrical system is required for towing compatibility with the military truck fleet as well as the commercial truck fleet (see 3.5).
(i) When the final painted color of the trailer, other than Forest Green, is required, the FED. STD. No. 595 color code number(s) of the color required, and the compatible marking color required, in lieu of black (see 3.8 and 3.9).
(j) Technical publications to be supplied (see 3.13).
(k) Time frame for the contractor to complete and submit the Technical Information Package (see 3.15).
(l) Time frame for the contractor to submit the First Article Demonstration Report (see 4.3).
(m) Type of packaging required (see 5.).



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6.3 First article model. Any changes or deviations of production trailer chassis from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish trailers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each trailer chassis (see 3.13).

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the trailer, covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.6).

6.6 Provisioning. The contracting officer should include provisioning requirements for BIIL, repair parts, and maintenance tools as necessary (including any special tools) and instructions regarding shipment of the trailers (see 3.13).

6.7 Government-furnished property. The contracting officer should arrange to furnish to the contractor the property as specified in 3.12.

6.8 Registration markings. The contracting officer should furnish a list of USA registration markings to be applied to the trailers (see 3.9).

6.9 Commercial users. The contracting officer should request the personnel contacts and company name and addresses of at least five civilian construction contractor users of the contractor's heavy hauler trailer product line being offered to the Government (see 3.1).

6.10 Vehicle weight classification. The contracting officer should arrange to furnish the contractor with vehicle classification blank forms and request the return of the completed forms to the Government. The technical data thus obtained should be utilized to determine the weight classification number for the trailer at GVWR and with the 30,000 pound payload capacity. The contractor should then be requested by the contracting officer to apply the weight classification number kit on each trailer. The use of either a contractor-furnished or Government-furnished sign kit conforming to MIL-S-40626 should be determined and furnished (see 3.18).



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6.11 Prime mover-trailer compatibility. The following prime mover references are provided for information:

| Prime Mover | Specification |
|---|---|
| Commercial Trucks | KKK-T-701 |
| CCE Dump Truck | MIL-T-52900 |
| CCE Truck Tractor | MIL-T-52901 |
| M44 Series Trucks, 2-1/2-Ton, Multifuel | MIL-T-45362 |
| M39 Series Trucks, 5-Ton | MIL-T-740 |
| M123 Series Trucks, 10-Ton | MIL-T-62012 |
| Rubber Tire Tractor | MIL-T-82118 |
| Rubber Tire Medium Tractor | MIL-T-52272 |
| Crawler Tractor | MIL-T-52270 |
| Crawler Tractor | MIL-T-52510 |
| Loader, Scoop Type, Pneumatic Tire | MIL-L-52385 |
| Loader, Scoop Type, Full Track | MIL-L-52431 |
| Loader, Scoop | MIL-L-52522 |

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 2330-A720



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US Army Mobility Equipment Research
and Development Command
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Ft. Belvoir, VA 22060

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-52479E(ME) Trailer, Flat Bed; Minimum 15 Ton Payload, Multi-Axle, Commercial Construction Equipment (CCE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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